

Exercises #11

[2*] 1. Read a sequence of double values into a vector. Think of each value as the distance between two cities along a given route. Compute and print the total distance (the sum of all distances). Find and print the smallest and greatest distance between two neighboring cities. Find and print the mean distance between two neighboring cities. For the smallest and greatest distances, write the separate functions called `smallest` and `greatest` with the following declarations:

```
double smallest(const std::vector<double>&);
```

```
double greatest(const std::vector<double>&);
```

Take two approaches:

- Use structured programming constructs like *for* and *if* statements to compute the smallest and greatest distances.
- Use generic algorithms *min_element* and *max_element* and replace your written code - the body of the above functions - with these functions.